

Growth-Flow Analytics **Customer Retention Strategy Optimization**

Technical Evaluation & Deployment Summary

Presented By: Ene Odu

August 8th, 2026

Background



Founded in 2010, **Growth Flow** is a global SaaS provider serving over 30,000 clients across North America, Europe, and Asia.



It provides an integrated CRM, marketing automation, and sales platform.



Despite achieving \$500M in annual revenue, the company is experiencing slowing customer retention rates.

Business Problem & Pain Points

01

Limited visibility into how retention changes across signup periods and subscription tiers

02

Missing time-based data on when churn happens

03

Inability to isolate high-risk behavior patterns or recognize high-performing retention segments.

04

Lack of timely, automated triggers to engage customers before they drop off.

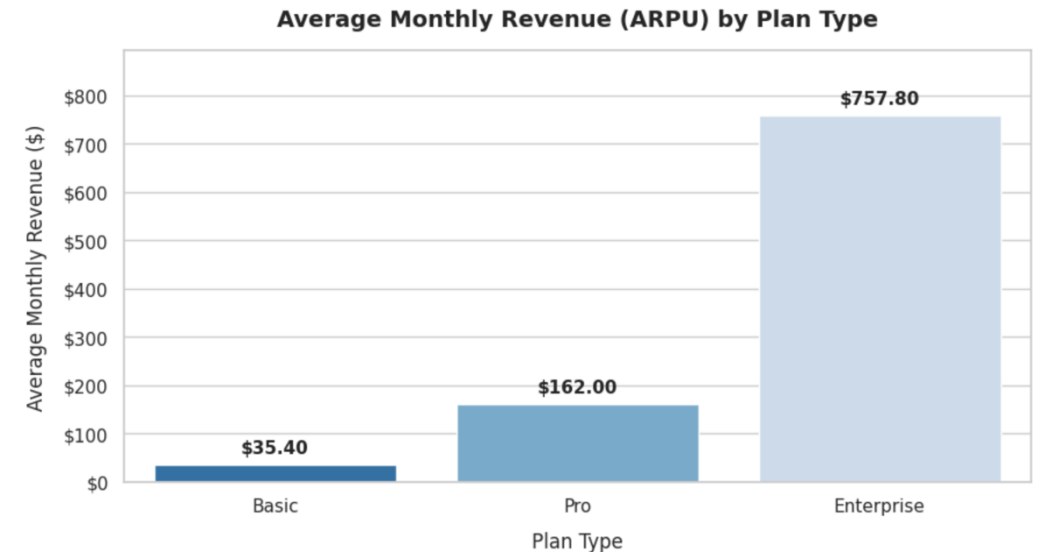
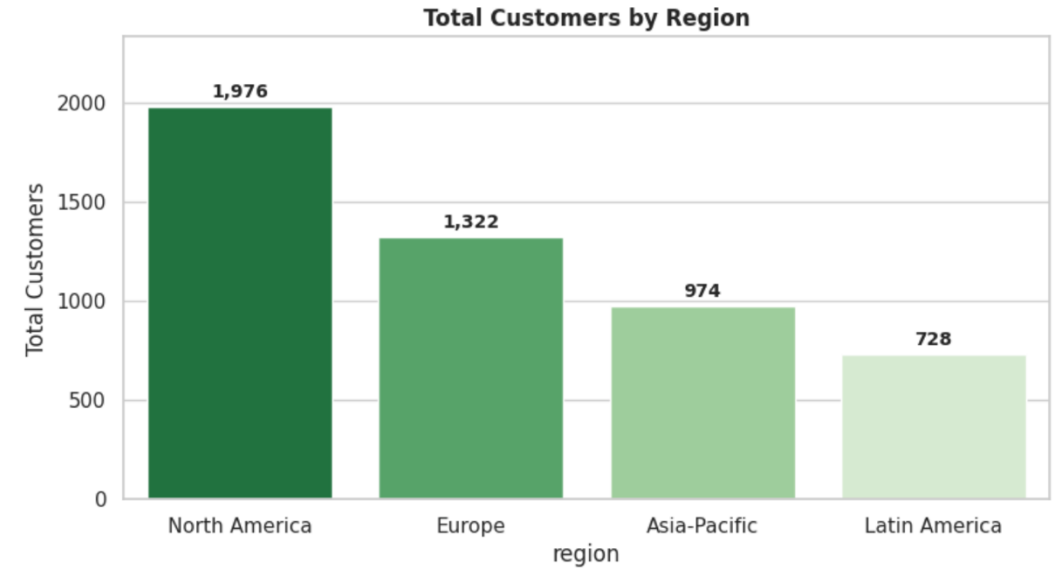
Data Scope

- 5,000 customer profiles (55,000 underlying event logs) collected over 24 months records contained in 5 databases.

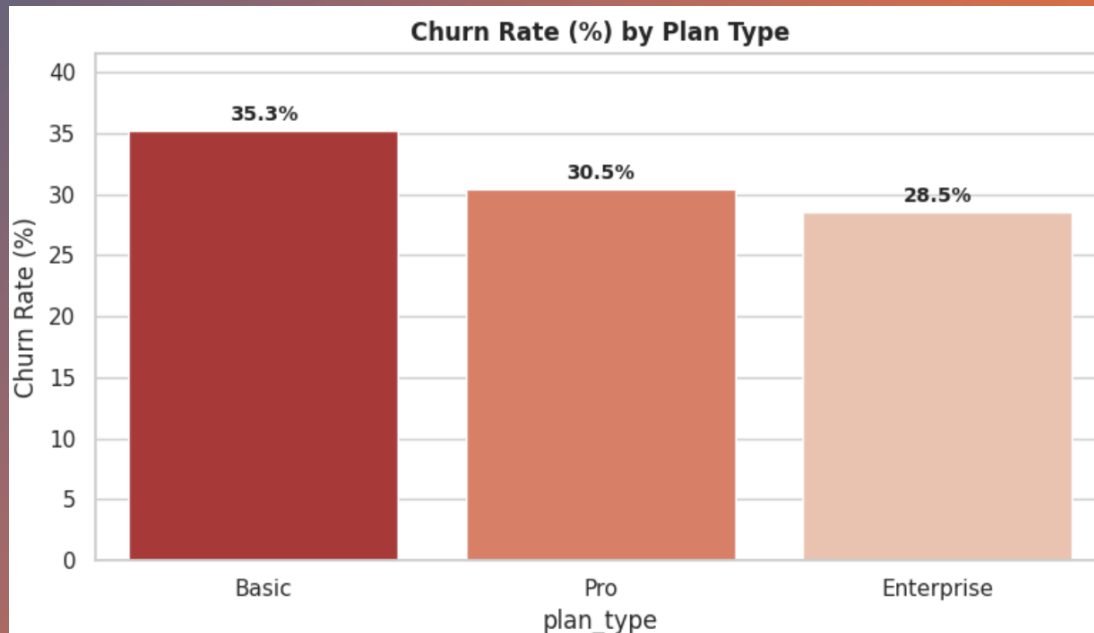
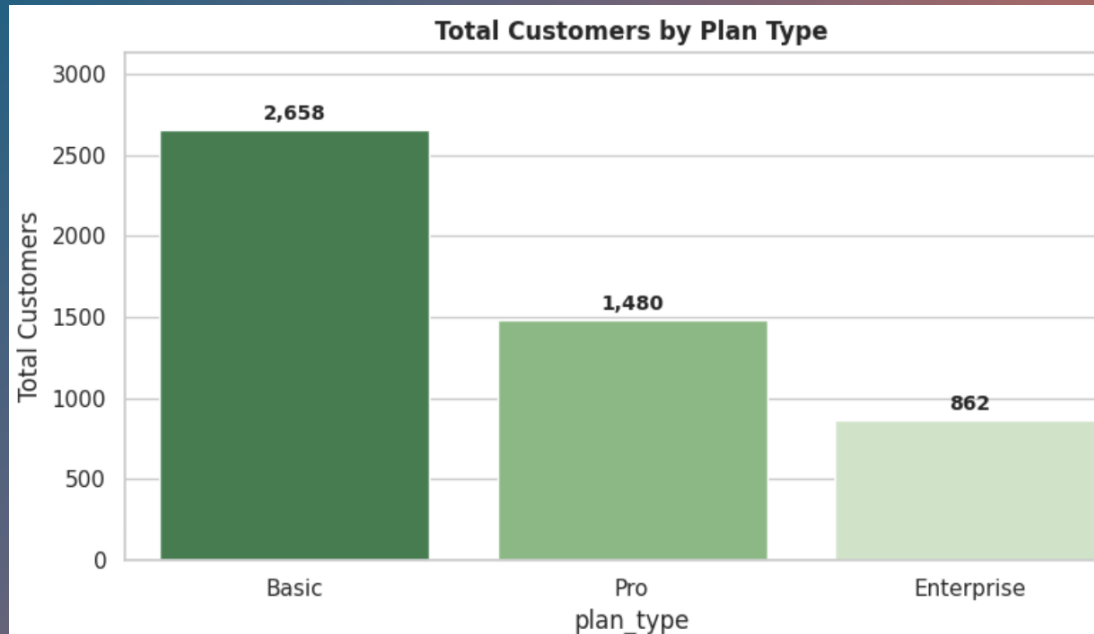
Dataset Table	Primary Role	Update Frequency
Customer Information	Base demographic & account attributes	Static Account Creation Monthly
Subscriptions	Core churn & retention tracking table	
Usage Data	Behavior & feature engagement tracking	
Feedback Data	Customer sentiment & health signals	
Engagement Data	Communication responsiveness metrics	

EDA Highlights

- **Data Completeness:** 100% complete data with zero missing values across all five database tables (55,808 total monthly records across 5,000 unique customers).
- **Customer Profile:** Primarily North American SMBs (60.24% of accounts) subscribing to Basic tier plans (53.16%).
- **Tier Pricing Expansion:** Significant revenue jumps exist across tiers:
 - Enterprise generates ~4.7x more revenue than Pro (\$758 vs \$162).
 - Pro generates ~4.6x more revenue than Basic (\$162 vs \$35),



Overview of Churn Problem



- Analysis of 5,000 customer profiles reveals an overall 32.7% churn rate and an Average Revenue Per User (ARPU) of \$193.25.
- Churn is not sudden; it is preceded by a structural drop in engagement.
- The Average Session Decay Ratio sits at 0.71 meaning customers drop 29% of their session depth prior to cancellation.
 - Executive Impact: This 29% drop provides an actionable 30 to 45 day window for Customer Success to intervene before revenue loss occurs.
- Churn results in \$265,876 in lost Monthly Recurring Revenue across 1,634 canceled accounts. While Basic plans account for 57.3% of cancellation volume, Enterprise cancellations account for 62.8% (\$166,975/month) of total financial loss.

Model Experimentation

- **Pipeline:** Executed Logistic Regression and Random Forest modeling after ingesting and merging records across all 5 source tables.
- **Dataset Scope:** Built a snapshot dataframe of 5,000 unique customer rows with 19 engineered features covering demographics, usage averages, sentiment, and pre-churn decay metrics.
- **Model Performance:** While Logistic Regression yielded slightly higher raw accuracy (67.9% vs. 64.8%), it failed operationally by predicting almost no churners. The Balanced Random Forest model resolved this, boosting Churn Recall from 8.3% to 48.6% and capturing nearly 6x more actual churners.

--- 1. DETAILED REPORT: LOGISTIC REGRESSION (BASELINE) ---

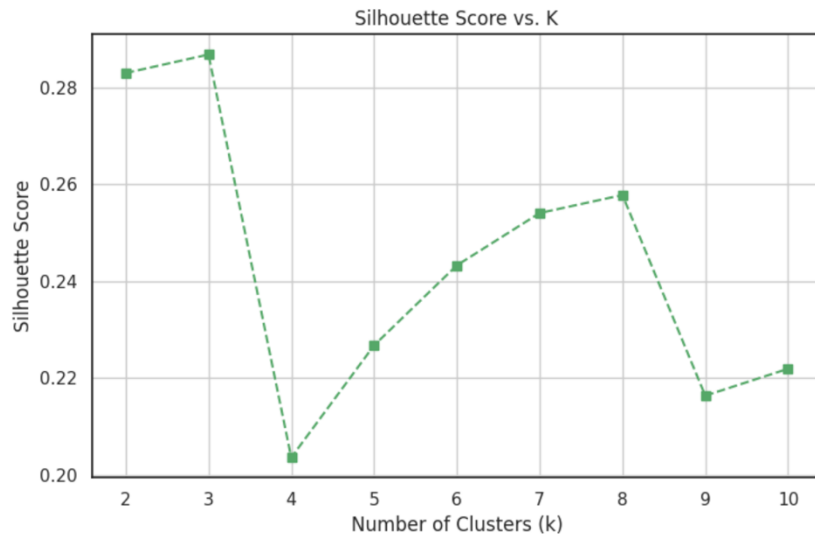
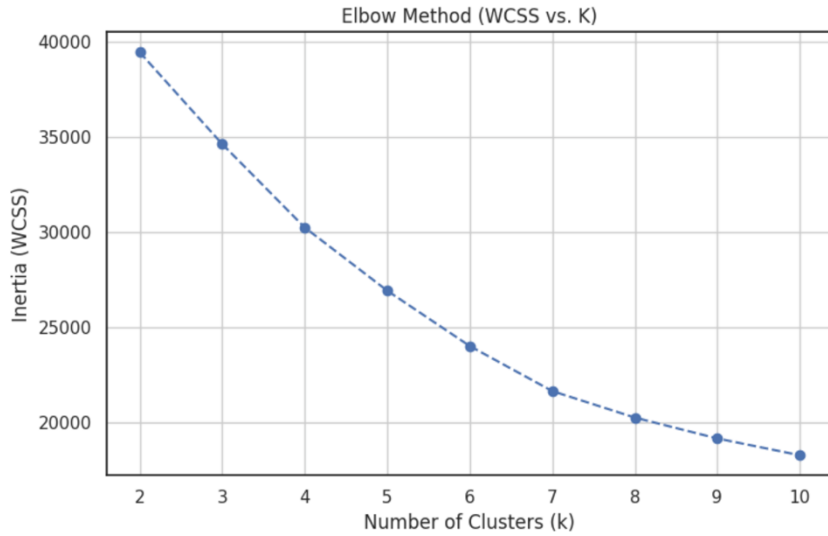
	precision	recall	f1-score	support
0	0.68	0.97	0.80	673
1	0.56	0.08	0.13	327
accuracy			0.68	1000
macro avg	0.62	0.52	0.47	1000
weighted avg	0.64	0.68	0.58	1000

--- 2. DETAILED REPORT: RANDOM FOREST (BALANCED) ---

	precision	recall	f1-score	support
0	0.75	0.71	0.73	673
1	0.46	0.51	0.49	327
accuracy			0.64	1000
macro avg	0.60	0.61	0.61	1000
weighted avg	0.65	0.64	0.65	1000

Customer Segmentation (Kmeans Clustering)

- Macro Tiers (k = 3 Clustering):
 - Cluster 0 (Power Champions - 16.8%): Large/Enterprise firms with high usage (90.52 uses/mon.) and \$505.69 ARPU.
 - Cluster 1 (Core SMB Base - 64.5%): Basic plans with low usage (18.30 uses/mon.) and \$109.47 ARPU.
 - Cluster 2 (Mid-Market Transition - 18.7%): Mid-sized businesses with steady adoption and \$201.61 ARPU.
- Root Cause of Churn (k = 8 Persona Insight):
 - Reveals distinct root causes of churn:
 - **Struggling On borders (Cluster 0):** Months 1–3 | Low usage | **37% Churn** (Early setup friction).
 - **At-Risk Long-Term Users (Cluster 1):** Month 12+ | Usage decay | **38% Churn** (Gradual disengagement).
 - **Engaged Newcomers (Cluster 6):** Month 4 | High email activity | **32% Churn** (Needs ongoing training to become Champions).





Model Deployment and Dashboard Development

- **Live Web App:** Created an interactive online dashboard using Streamlit, Python, and Plotly instead of relying on static charts.
 - **Fast Performance:** Smart data caching automatically groups customers into 24-month tiers instantly without slowing down when users click around.
 - **Reliable Data Loading:** Added backup loading rules so the app seamlessly reads data without crashing, even if a file is missing.
-

Strategic Recommendations

- **Early Warning Triggers (0–30 Days):** Automatically flag accounts when engagement drops by 20% (Session Decay Ratio < 0.80) to prompt re-engagement before churn occurs.
- **Onboarding Redesign (Months 0–3):** Use product tours, feature checklists, and automated emails in the first 7 days to help Basic and Pro users build lasting habits.
- **Re-engage low-usage stable users around Month 12** by triggering automated check-ins, feature training, and personalized re-boarding.
- **Enterprise Renewal Playbook (Months 13–18):** Hold Executive Business Reviews 90 days before contract expiration for high-value accounts showing disengagement.



Next Steps



TRANSITION BATCH CSV TRANSFORMATIONS
TO AUTOMATED SQL PIPELINES FOR REAL-
TIME FEATURE EXTRACTION.



INCORPORATE PROPORTIONAL HAZARDS
MODELING TO ESTIMATE PRECISE CHURN
PROBABILITY CURVES AT SPECIFIC TENURE
DAY.



IMPLEMENT AUTOMATED TRACKING ON
STREAMLIT CLOUD TO MONITOR MODEL
PERFORMANCE DRIFT AS CUSTOMER USAGE
PATTERNS EVOLVE



Questions